

LAB OBSERVATION

EXPERIMENT-15

Implement probabilistic context-free grammar parsing for a sentence using python.

CODE:

```
[S] 1 import nltk
    2 from nltk import PCFG
    3 from nltk.parse import ViterbiParser
    4
    5 grammar = PCFG.fromstring("""
    6 S -> NP VP [1.0]
    7 NP -> 'John' [0.5] | 'Mary' [0.5]
    8 VP -> V NP [1.0]
    9 V -> 'likes' [1.0]
   10 """)
   11
   12 parser = ViterbiParser(grammar)
   13
   14 sentence = input("Enter sentence: ").split()
   15
   16 for tree in parser.parse(sentence):
   17     print(tree)
```

Variables Terminal 2:11 PM Python 3

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OUTPUT:

```
... Enter sentence: John likes Mary
(S (NP John) (VP (V likes) (NP Mary))) (p=0.25)
```

Variables Terminal 2:11 PM Python 3

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